

NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC

Subject: Engineering

Questions: 37

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AI-generated PYQs: 0

1. [UNKNOWN] paper-i

Q2. (a) What are rainfall Depth-Area-Duration (DAD) curves? Write down the step-by-step procedure for drawing Depth-Area-Duration curves. 10 (b) Explain in brief, different types of erosion and their causes. 10 (c) Define Water Harvesting. Write the components and sequential steps for runoff water harvesting systems in the dry regions of India with suitable sketches. Also write the limitations of water harvesting systems. 10 (d) Design a waste weir using the following data: 10 Cumulative catchment area: 35 ha Flood depth over the crest of weir: 0-6 m Intensity of rainfall: 50 mm/hr Runoff coefficient: 0-5

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2. [UNKNOWN] paper-i

Q2. (a) A uniform T-section beam is 100 mm wide and 150 mm deep with a flange thickness of 25 mm and a web thickness of 12 mm. If the limiting bending stresses for the material of the beam are 80 MN/m² in compression and 160 MN/m² in tension, find the maximum U.D.L. that the beam can carry over a simply supported span of 5 m. 10
FSI-P-CVLE 2 ===== PAGE 3 ===== A steel column consisting of ISMB 350 is effectively restrained at mid-height by a bracing member in Y-Y direction, but is free to move in Z-Z direction and both the ends of the column are pinned. Its unsupported length is 6 m. Determine its axial load carrying capacity at service loads, using the limit state design of IS: 800 — 2007. / 7mo Design compressive strength $f_{cd} = \frac{f_y}{\gamma_{m1}} \left[1 - \alpha (X - 0.2) + A_2 \right]$ If $f_y = 250$ N/mm² $\gamma_{m1} = 1.10$ and $\gamma_{m2} = 1.5$ $E = 2 \times 10^5$ N/mm² Minimum radius of gyration for ISMB 350 is given below: $r_{yy} = 28.4$ mm $r_{xx} = 142.9$ mm Area of cross-section of the ISMB 350 is 6671 mm² In the framed structure shown in the figure below, if the length of all the members is the same, determine the force in the member 'DE' using the Virtual Work Method. 200N 200 N Rb 15 10
FSI-P-CVLE 3 ===== PAGE 4 ===== (d) A railway wagon weighs 0-25 MN when empty and 0.6 MN when loaded. The springs of the wagon get compressed by 8 cm when it is full. Determine the natural period of vibration when the wagon is empty and when it is loaded. 5

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3. [UNKNOWN] paper-i

Q2. SECTION A A slider-crank reciprocating mechanism has the following data: Radius of the crank = 480 mm Length of the connecting rod = 1600 mm Angular velocity of the crank = 20 rad/s Find the velocity and acceleration of the piston for the crank position of 45° from the inner dead centre. Two pulleys having diameters 640 mm and 480 mm, are used to connect two parallel shafts by a crossed belt drive. The distance between the centre lines of the shafts is 3 m. Determine how much the length of the belt should be changed, if it is desired to alter the direction of rotation of the driven shaft. A rod of 1 m length is kept at a temperature of 30°C. Find the expansion of the rod when the temperature is raised to 80°C. If this expansion is prevented, find the stress induced in the material of the rod Take $E = 100$ GPa and $\alpha = 0.000012/^\circ\text{C}$. (d) The state of stress at a point in a loaded piece of material is given by $\sigma_x = 85$ kPa $\sigma_y = -40$ kPa and $\tau = +50$ l(Pa. y XY — Find the magnitudes of the principal stresses. Also find the magnitude of the maximum shear stress and the plane on which this acts. In a simple cubic crystal, draw the following planes: (110) (111) (a) (i) State the law of gearing. Also state the condition to meet the above requirement. Name the forms of teeth profiles normally used to avoid interference between the mating gears. (ii) A pair of pinion and gear of 20° pressure angle involute profiles mesh externally and provide a velocity ratio of 3. The addendum is equal to 112 of module. Determine the minimum number of teeth on each to avoid interference. 5+10 8 8 8 8 FSI-P-MCHE 2 ===== PAGE 3 ===== A thick cylinder is made of 6 cm internal diameter. It is subjected to an internal pressure of 50 MPa. If the maximum tensile stress is limited to 100 MPa, find the thickness required. Also, show the variations of hoop and radial stresses across the thickness of the cylinder. 15 Draw a neat and clean iron-carbon equilibrium diagram and label its various features. 10

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4. [UNKNOWN] paper-i

Q3. (a) Write the step-by-step procedure for planning and design of a graded bunding. 10 (b) Design a grassed waterway of parabolic shape to carry a flow of $2-6 \text{ m}^3/\text{s}$, down a slope of 3%. The waterway has a good stand of grass and velocity of $1-75 \text{ m/s}$ can be allowed. Assume the value of Manning's roughness coefficient 'n' as 0.04. 10 (c) Discuss the functions, limitations and design features of chute spillways. 10 (d) What are the different types of farm ponds and their essential design requirements? 10 FSI-P-AGRE 2 SECTION A ===== PAGE 3 =====

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5. [UNKNOWN] paper-i

Q3. (a) A welded bracket connects a plate to the column flanges as shown in the figure. Determine the size of weld if the allowable stress in the weld is 110 N/mm^2 . 15 450 mm (b) A simply supported beam of span 8 m is subjected to a series of rolling loads as shown. Determine the position of loads and the absolute maximum Bending Moment. 15 80 kN 100 kN 70 kN 100 kN m If 2-4 m (c) A beam with a rectangular cross-section of size 250 mm wide and 500 mm deep is prestressed by a force of 500 kN using 8 Nos. of 7 mm ϕ steel cables located at an eccentricity of 80 mm. Find the loss of prestress due to creep of concrete. Grade of concrete is M 40. Coefficient of creep is taken as F_8 . Modulus of Elasticity of steel E_s is $2 \times 10^5 \text{ N/mm}^2$ Assume the stress at transfer is 80%. 10 FSIT-CVLE 4 ===== PAGE 5 =====

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6. [UNKNOWN] paper-i

Q3. (a) A single cylinder reciprocating engine has the following data: Mass of reciprocating parts = 40 kg Mass of revolving parts = 36 kg at crank radius Speed of crank = 150 rpm Stroke length of reciprocating parts = 350 mm All the revolving parts and 60% of the reciprocating parts need to be balanced. Determine: Balancing mass required at a radius of 340 mm Unbalanced force when the crank has turned 45° from the top dead centre 10 (b) A machine mounted on 4 springs and fitted with a dashpot has a mass of 120 kg. The stiffness of each spring is 18 N/mm . The amplitude of vibrations reduces from 33.75 mm to 6 mm in two complete oscillations. Assuming that the damping force varies as the velocity, determine damping coefficient, and ratio of frequencies of damped and undamped vibrations. 10 (c) A uniform beam ($I = 7,800 \text{ cm}^4$) is 6 m long and carries a central concentrated load of 60 kN. Taking $E = 210 \text{ GPa}$, calculate the deflection under the load if the beam is built-in at one end and simply supported at the other end. 10 (d) Draw the heating cycle on Temperature — Time scale of the following processes: 10 (i) Process annealing or Subcritical annealing Spheroidize annealing Austempering FSI-P-MCHE 3 ===== PAGE 4 =====

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7. [UNKNOWN] paper-i

Q4. (a) Discuss the principles of GIS and remote sensing and their applications in watershed development and management. 10 (b) Write short notes on the following : 10 (i) Universal Soil Loss Equation (ii) Catchment Area Treatment (iii) Agronomic Erosion Control Measures (iv) Functions and Planning of Terraces (v) Elements of Flood Routing (c) It is planned to construct bench terraces on a 20% hill slope. If the vertical interval is 2 m, determine ; 10 (i) Length per hectare (ii) Earthwork (iii) Area lost, both for vertical cut and batter slope of 1 : 1. (The cut should be equal to fill) (d) Discuss different methods of Gully Control. Explain the steps followed in the design of gully control structures. 10 FSI-P-AGRE ===== PAGE 4 =====

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8. [UNKNOWN] paper-i

Q4. (a) A continuous beam ABC is loaded as shown in the figure below. Determine the required Plastic Moment (M_p) of the beam section if the load factor is 3-2. 15 60 kN 90 kN 5 kN/m M_p D $2M_p$ E 12m A' 8m 8m 8m A masonry retaining wall, trapezoidal section is 1.2 m wide at the top. Its earth retaining face is vertical and smooth. The retained earth, having density of 16 kN/m^3 and angle of shearing resistance of 30° , is level with the top of the wall. Height of the wall is 6.9 m. The density of masonry is 19.5 kN/m^3 . Calculate the maximum bottom width of the wall, so that no tension is to be developed at the base. Also calculate the maximum base pressure at this width. 15 Two ISA 90 x 90 x 8 mm carry a load of 180 kN and are placed back to back through 8 mm gusset plate. The permissible shear stress is 110 N/mm^2 and bearing stress is 250 N/mm^2 . Design the riveted connection and show the arrangement with a neat sketch. 10 ESP CVLE 5 ===== PAGE 6 ===== SECTION B

9. [UNKNOWN] paper-i

Q4. (a) Each arm of a Porter governor is 200 mm long and is pivoted on the axis of the governor. The radii of rotation of the balls at the maximum and minimum speeds are 160 mm and 120 mm respectively. The mass of each ball is 4 kg and the mass of sleeve is 24 kg. Determine the range of speed of this governor. Also find the range of speed, if the friction at the sleeve is 18 N. 10 A shaft is required to transmit 40 kW at 300 r.p.m. The maximum torque may be 1-5 times the mean torque. The allowable twist is 10 per m length of the shaft. Determine the diameter of the solid shaft. /0 A column of 6 m length is fixed at both ends. It is of rectangular cross-section of 20 cm x 10 cm. Determine the Euler's crippling load on the column, if $E = 200 \text{ GPa}$. 10 Determine the composition, in atom percent, of an alloy that consists of 97 wt% aluminium and 3 wt% copper. /0 F5I-P-MCHE 4 ===== PAGE 5 ===== SECTION B

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10. [UNKNOWN] paper-i

Q5. Answer the following : 8y.5-40 (a) Differentiate between the following : 8 (i) Water requirement and Irrigation requirement (ii) Crop water use efficiency and Field water use efficiency (iii) Evaporation and Consumptive use (iv) Confined and Unconfined aquifers (b) Discuss the procedure to develop soil moisture characteristic curve and its critical points. 8 (c) Discuss Darcy's law of soil water flow with a definition sketch. In what flow conditions is it applicable ? 8 (d) What is Cavitation ? How does it affect the performance of a pumping system ? How is it prevented in a centrifugal pump ? 8 (e) A field soil sample prior to being disturbed has a volume of 82 cm^3 . The sample weighed 125 grams. After drying at 105°C for 24 hours, the dry soil sample weighs 100 grams. What is the soil moisture content on weight basis and volume basis ? What depth of water must be applied to increase the volumetric water content of the top 1 m of soil to 0-40 ? Assume the density of water as 1000 gm/cm^3 . 8

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11. [UNKNOWN] paper-i

Q5. (a) The velocity potential for a two-dimensional flow is $\phi = x(2y - 1)$. Determine the value of stream function at a point P(4, 5). Comment whether the flow is rotational or irrotational. Also determine the velocity at the point P(4, 5). 8 (b) A three-cylinder car has pistons of 75 mm and cylinders of 75 mm Find the percentage change in force required to drive the piston, when the lubricant warms from 25°C to 100°C . The dynamic viscosity of the lubricant at 25°C is 2 Ns/m^2 and at 100°C , it is 0.4 Ns/m^2 . 8 (c) A soil sample is tested in the laboratory for classification. The test results are as follows : Percentage passing 4.75 mm IS sieve = 58 Percentage passing 75 ft IS sieve = 11 The particle size $D_{60} = 6.3 \text{ mm}$ $D_{30} = 0.5 \text{ mm}$ $U_{0.7} = 0.07 \text{ mm}$ Liquid limit = 42 Plastic limit = 21 Classify the soil. 8 (d) A constant head permeability test was run on a sand sample 400 mm long and 3000 mm^2 in area. When the loss of head was 900 mm, the quantity of water collected in 3 minutes was 400 ml. Determine the coefficient of permeability of the soil. If the specific gravity of grains is 2.65 and the dry mass of the sample is 1.9 kg, find the void ratio of the sample. 8 (e) Compare the salient features of Standard Penetration Test and Plate Load Test. 8 FSI-P-CVLE 6 ===== PAGE 7 =====

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12. [UNKNOWN] paper-i

Q5. (a) Explain the variables influencing machinability. How is machinability assessed? 8 (b) Determine the material removal rate and the electrode feed rate in the electrochemical machining of an iron surface that is $10 \text{ mm} \times 10 \text{ mm}$ in cross-section, using NaCl in water as electrolyte. The gap between the tool and workpiece is 0.25 mm. The supply voltage is 12 V DC. The specific resistance of the electrolyte is $4 \times 10^{-2} \text{ cm}$. For iron, atomic weight = 55.85, valency $z = 2$ and density $\rho_a = 7860 \text{ kg/m}^3$. 8 (c) Enlist the areas of project monitoring. What are the steps for project monitoring? 8 (d) An analysis of a company reveals the following sales and cost information: Current capacity = 1,00,000 units At current level of operations, its margin of safety is 5% of its break-even point, whereas contribution margin P/V ratio is 25% and unutilised capacity is 10,000 units. For the sale price of 40 per unit, determine the following: 8 Break-even point in sales volume Fixed costs Variable costs per unit Margin of safety in units (e) Define ROM, PROM, EPROM, EEPROM and RAM. 8

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13. [UNKNOWN] paper-i

Q6. (a) Draw a neat sketch showing the basic components of a drip irrigation system. Also describe the functions of each of them in brief. 10 (b) Discuss the important aquifer characteristics influencing yields of wells. 10 (c) Discuss sprinkler irrigation uniformity and its measures and describe the procedure to estimate it in the field. 10 (d) Given: SECTION B Climate Hot humid Soil Fine sandy loam Wind speed 7 km/h Crop Wheat Effective root zone depth 1-5 m Water available in top soil {1-0 m} 167 mm/m Water available in top soil (0-5 m) 183 mm/m Irrigation application efficiency 80% E C , 2-0 dS/m ECg {for 10% yield reduction} 3-4 dS/m Area 10 ha Peak rate of water removal 7-6 mm/day Management allowed deficit (MAD) 50% Operating hours per day 12 hours Find the Net and Gross depth of irrigation, Leaching requirement. Irrigation interval and Sprinkler system capacity. 10 FSI-P-AGRE ===== PAGE 5 =====

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14. [UNKNOWN] paper-i

Q6. (a) Find the hydraulic efficiency of a Felton turbine operating under a head of 350 m, with a discharge of 15 m³/s. 15 The other data are: Speed ratio $K_u = 0.46$ Coefficient of velocity $C_s = 0.985 = 0.9 V_{ri}$ and $(I) = 20^\circ$ A raft of size 4 m x 4 m carries a uniform load of 200 kN/m². Find the stress at 6 m below the raft centre considering both geostatic stress and induced stress. Use point load approximation with four equivalent point loads. Take 20 kN/m³. 15 There is a bed of compressible clay of 3 m thickness with pervious sand on top and impervious rock at the bottom. In a consolidation test on an undisturbed specimen of this clay, 90% settlement is reached in 6 hours. The specimen was 20 mm thick. Estimate the time in years for the building, founded over this deposit, to reach 90% of its final settlement. 10

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15. [UNKNOWN] paper-i

Q6. (a) During orthogonal turning operation of a medium carbon steel rod with a carbide tool having orthogonal rake angle 100, the following observations were made: Width of the chip = 6 mm Uncut chip thickness = 0.1 mm Chip thickness ratio = 0.4 Horizontal component of cutting force = 1300 N Vertical component of cutting force = 1650 N Determine the component of cutting force along the rake face and shear plane. Also find out coefficient of friction, resultant force and shear stress. 15 FSI-P-MCHE 5 ===== PAGE 6 ===== (b) Explain the features and policy guidelines for ABC analysis. A manufacturer has to supply his customer 3600 units of his product per year. Shortages are not permitted. Inventory carrying cost amounts to 1.2 per unit per annum. The set-up cost per run is 80. Compute the following : 15 Economic order quantity Optimum number of orders per annum Average annual inventory cost (minimum) Optimum period of supply per optimum order (c) Draw a flow chart for computation of simple interest for 3 sets of p, n, r. 10

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16. [UNKNOWN] paper-i

Q7. (a) Differentiate between ESP and SAR. Derive the relationship between these two indices. What are salt affected soils ? Explain different reclamation measures to manage salt affected soils. 10 (b) A 25 cm diameter well is pumped at a uniform rate of 50 Z/s. The drawdown observed at 1 m and 100 m distances from the centre of the well are 8 m and 40 m respectively. Determine the hydraulic conductivity of the water bearing strata. Assume the thickness of the saturated part of the aquifer as 25 m. 10 (c) Discuss the Pumping Test of wells. Explain the step-by-step procedure adopted for conducting pumping tests of wells. 10 (d) What are submersible pumps ? Explain the complete procedure to install a submersible pump in a bore well including initial preparations, installation and testing. 10

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17. [UNKNOWN] paper-i

Q7. (a) What is hydraulic jump 9 Explain the uses of hydraulic jump in brief. Derive the equation for sequent depth ratio, for a hydraulic jump in a horizontal rectangular channel. What are the assumptions used in deriving the sequent depth ratio? 10 A rectangular channel is 4.0 m wide and conveys a discharge of 20 m³/s at a depth of 2.2 m. It is proposed to reduce the width of the channel at a hydraulic structure. Assuming the transition to be horizontal and the flow to be frictionless, determine the water surface elevations upstream and downstream of the constriction when the constricted width is 2.8 m. 15 A gravity retaining wall retains 12 m of backfill, $\gamma = 18$ kN/m³, $\phi = 30^\circ$ with a uniform horizontal

surface. Assuming the wall interface to be vertical, determine the magnitude and point of application of total active pressure. If the water table is 3 m below the ground surface, how far do the magnitude and point of application of active pressure change ? Assume submerged unit weight of soil as 11 kN/m³. 15 FSI-P-CVLE 7 ===== PAGE 8 ===== QS.

(a) Two reservoirs 10 km apart have a difference of water level of 25 m and are connected together by a pipeline. The Pipeline consists of a 6 km single pipe feeding a junction from which two pipes run in parallel for the remaining length of 4 km to the lower reservoir. If the same size of pipe is used throughout, what should be the diameter for the pipe so that velocity may not exceed 1.25 m/s ? Take the friction coefficient $f = 0.03$. 15 For the following data, compute the settlement of the group assuming the load to be transferred at two-thirds the length of a pile : Depth of clay layer = 16 m, load on piles = 2000 kN, length of piles = 10.5 m, diameter of a pile = 0.45 m, spacing of piles = 1.25 m c/c, number of piles = 16, initial void ratio = 0.80, $C_e = 0.45$ and unit weight of soil = 16.5 kN/m³. Take size of pile cap 5.65 x 5.65 m. 15 A pipe of diameter 1.5 m is required to transport an oil of specific gravity 0.9 and viscosity 3×10^{-2} poise at the rate of 3000 litres/s. Tests were conducted on a 15 cm diameter pipe using water at 20°C. Find the velocity and rate of flow in the model. Viscosity of water at 20°C is 0.01 poise. 10 FSI-P-CVLE 8

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18. [UNKNOWN] paper-i

Q7. (a) While turning a mild steel rod with a HSS cutting tool at a feed of 0.1 mm/rev, and depth of cut of 3 mm, the following observations were made: Cutting velocity V , 20 30 min Tool life T , min 70 40 Determine the cutting velocity, if desired tool life is 60 minutes. 10 A cylindrical pipe of mild steel with inside diameter 60 mm and thickness 2.5 mm is to be reduced down to 48 mm and thickness 1.75 mm. Consider die angle as 40° and coefficient of friction is 0.1. Compare the pipe drawing force on cylindrical plug and movable mandrel. 10 A company intends to buy a machine having a capacity to produce 1,70,000 quality parts per annum. The machine constitutes a part of the total product line. The system efficiency of the product line is 85%. Find the system capacity per hour. The time required to produce each part is 100 seconds and the machine works for 2000 hours per year. If the utilization of the machine is 60% and the efficiency of the machine is 90%, compute the output of the machine per hour. Calculate the number of machines required. 10 FSI-P-MCHE 6 ===== PAGE 7 ===== (d) A linear programming problem is given subjected to the constraints $53(1 + x_2 \leq 10x_1 + x_2 \leq 6x_1 + 4x_2 \leq 12x_1, x_2 \geq 0$ For minimization of the function $z = 3(1 + 2x_2)$, plot each constraint on a graph paper and show the feasible region. 10

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19. [UNKNOWN] paper-i

Q8. (a) Discuss various types of poly houses (low to high cost) used for cultivation of high value crops. Explain the working principle of fan and pad evaporative cooling systems. 10 (b) Discuss dairy bams, their types and their components. Draw a model layout of a dairy farm for a 20 cow/buffalo unit. 10 (c) Discuss the purpose, benefits and types of subsurface drainage systems. Explain the design of pipe drainage systems used for large irrigation projects. 10 (d) What is a Farmstead ? What factors govern the location of the farmstead on the farm ? 10 FSI-P-AGRE

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20. [UNKNOWN] paper-i

Q8. (a) (i) Determine the dimensions of 'Go' and 'No Go' gauges of a plug gauge for checking 75 ± 0.05 mm diameter holes. Show the dimensions using Bilateral and Unilateral systems. Consider gauge maker's tolerance as 10% of work tolerance. 10 (ii) What information can you obtain from the following surface roughness representation? 5 (b) A small project is composed of 7 activities whose time estimates are listed in the table below. Activities are identified by their beginning (i) and ending (j) node numbers. Activity Estimated duration (in weeks) (i — j) Optimistic Most likely Pessimistic 1 — 2 1 1 7 1 — 3 1 4 7 1 — 4 2 2 8 2 — 5 1 1 1 3 — 5 2 5 14 4 — 6 2 5 8 5 — 6 3 6 15 Draw the network diagram and critical path. Find the expected duration and variance for each activity. Find the expected project length. 15 (c) Write a C-program using Array to find out the average marks obtained by a class of 50 students in a test conducted. 10 FSI-P-MCHE 7

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21. [UNKNOWN] paper-ii

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22. [UNKNOWN] paper-ii

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23. [UNKNOWN] paper-ii

1. (b) Calculate the approximate Grashof number and state if the flow is laminar or turbulent for the following: A central heating radiator, 0.6 m high with a surface temperature of 75°C in a room at 18°C, $p = 12 \text{ kg/m}^3$, $Pr = 0.72$ and $p = 1.8 \times 10^{-5} \text{ kg/ms}$. A horizontal oil sump with a surface temperature of 40°C, 0.4 m long and 0.2 m wide containing oil at 75°C. Take $p = 854 \text{ kg/m}^3$, $Pr = 546$, $f_3 = 0.7 \times 10^{-3}$ and $\mu = 3.56 \times 10^{-2} \text{ kg/ms}$. 10

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24. [UNKNOWN] paper-ii

1. (c) Discuss the causes, effects and the actions to be taken to remove/reduce the following emissions from I.C. engines: Oxides of Nitrogen (NO) Smoke Carbon Monoxide (CO) Unburned Hydrocarbons (HC) Sulphur Oxides (SO₂) 10

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25. [UNKNOWN] paper-ii

1. (d) A chimney has a height of 100 metres. For the maximum discharge condition, calculate the temperature of the chimney gases and the draught produced if the air supplied per kg of fuel is 18 kg. Also determine the efficiency of this chimney as an instrument for creating the draught, if the temperature of chimney gases in artificial draught system is limited to 120°C. Take the boiler house temperature as 40°C and the specific heat of flue gases as 1.005 kJ/kg K. 10
1.(a) fsi-d-mche 2 ===== PAGE 3 ===== Temperature CP j.t K Pr °C J/kg K N.s/m² W/mK 100 2250 1.41x10⁻² 0.137
300 80 2131 3.25x10⁻² 0.138 546 60 2035 8-36x10⁻² 0-141 1205 Temperature Co 1.1 Pr °C J/kg K N.s/m² W/mK 30
4178 769 x 10⁻⁶ 0.620 5-20 35 4178 725x10⁻⁶ 0-625 4-85 40 4179 631x10⁻⁶ 0-634 4.16 • 2.(a) A four-cylinder
four-stroke S.I. engine with 80 mm bore and 90 mm stroke, runs at 4000 rpm and uses a fuel having 84% carbon and 16% hydrogen by mass. The volumetric efficiency of the engine at that speed is 80%. The ambient conditions are :
pressure = 1.0 bar, temperature = 25°C. The depression at the venturi throat is 0.06 bar The actual quantity of air supplied is 0.95 of the stoichiometric value. Calculate the fuel flow rate, the air velocity at the throat and the throat diameter. Take : R(air) = 287 J/(kg K), R(fuel vapour) = 98 J/(kg K) Cp = 1005 J/(kg K), $\gamma = 1.4$ and coefficient of discharge at venturi throat as 1. 20 2.(b) A counterflow, concentric tube heat exchanger is used to cool the lubricating oil for a large industrial gas turbine engine. The flow rate of cooling water through the inner tube ($D_i = 25 \text{ mm}$) is 0.2 kg/s, while the flow rate of oil through the outer annulus ($D_o = 45 \text{ mm}$) is 0.1 kg/s. The oil and water enter at temperature of 100 and 30°C respectively. How long must the tube be made if outlet temperature of the oil is to be 60°C? State assumptions made, if any. Comment on the magnitude of length of tube. 20 Properties of Oil Properties of Water
Nusselt number for fully developed laminar flow in a circular tube annulus with one surface insulated and the other at constant temperature: D_i/D_o Nu_i Nu_o D_i/D_o Nu_i Nu_o 0 3.66 0.25 7.37 4.23 0.05 17.46 4.06 0.50 5.74 4.43 0.10 11-56
4.11 1.00 4.86 4-86 3 fsi-d-mche ===== PAGE 4 ===== --

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26. [UNKNOWN] paper-ii

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27. [UNKNOWN] paper-ii

3. (a) The output of a three stage gas turbine is 30 MW at the shaft coupling at an entry temperature of 1500K. The overall pressure ratio across the turbine is 11 and efficiency 88%. If the pressure ratio of each stage is same, determine : pressure ratio of each stage polytropic efficiency the mass flow rate the efficiency and power of each stage The properties of the working medium are the same as of air ($\gamma = 1.4$, $C_p = 1.005 \text{ kJ/kg K}$). Assume an efficiency of 91% to take into account shaft losses due to disc and bearing friction. 20

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28. [UNKNOWN] paper-ii

3. (b) A retail shop located in a city at 30° N latitude has the loads as given below:. Room Sensible heat — 58.15 kW
Room Latent heat — 14-54 kW The summer outside and inside design conditions are : Outside : 40°C DB , 27°C WB
Inside : 25°C DB , 50% RH 70 m³/min of ventilation air is used. Determine the following: Ventilation load Grand total heat
Effective sensible heat factor Apparatus dew point Dehumidified air quantity Condition of air entering and leaving
apparatus Given a choice, what bypass factor (BF) would you choose from 0.05, 0.04, 0.15, 0.20. Give justification for
selection of BF. Solve the problem using BF of 0.15. Use of Psychrometric chart is permitted. 20 fsi-d-mche 4 '1 =====
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29. [UNKNOWN] paper-ii

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30. [UNKNOWN] paper-ii

4. (a) Describe the working principle of Jerk type injection pump with the help of neat diagram. Show the position of the helix for various load conditions. 20 (b) In a reversed Camot refrigerator system of 1TR cooling capacity running on perfect gas, heat is absorbed at -10°C and rejected at 50°C . Find the states at all the points of the cycle, mass flow rate, volume flow rates and COP. The maximum pressure ratio is 5 and the pressure at inlet to the isentropic process is standard atmospheric pressure. Take $C_p = 1.005 \text{ kJ/kg K}$, $R = 0.287 \text{ kJ/kg K}$ and $\gamma = 1.4$. Plot the cycle on $p-v$ and $T-s$ diagrams. Comment on the area of $p-v$ diagram. 20 SECTION 'W (a) A 3 cm OD pipe is to be covered with two layers of insulation, each having a thickness of 2.5 cm. The average thermal conductivity of one insulation is five times that of the other. Determine the percentage decrease in heat transfer if the better insulating material is next to the pipe than if it is the outer layer. Assume that the outside and inside surface temperatures of the composite insulation are fixed. 10

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31. [UNKNOWN] paper-ii

5. (b) Air is compressed steadily by a reversible compressor from an inlet state of 100 kPa and 300 K to an exit pressure of 900 kPa. Determine the compressor work per unit mass for (i) Isentropic compression with $\gamma = 1.4$; Polytropic compression with $n = 1.3$; (iii) Isothermal compression and (iv) Ideal two-stage 'compression with intercooling with a polytropic exponent of 1.3. 10

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32. [UNKNOWN] paper-ii

5. (c) Discuss the effect of the following variables on ignition lag: Nature of fuel and air/fuel ratio Initial temperature and pressure Spark timing Turbulence and engine speed Gap between electrodes of the spark plug 10 5.(d)

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33. [UNKNOWN] paper-ii

6. (a) Determine the pressure ratio developed and the power required to drive a centrifugal compressor (impeller diameter = 45 cm) running at 7200 rpm. Take zero swirl at entry and $T_{01} = 288 \text{ K}$. Assume isentropic flow with no shock, and radially tipped impeller blades. Take $C_p = 1.005 \text{ kJ/kg K}$ and $\gamma = 1.4$. 10 A food compartment of a refrigerator

is maintained at 4°C by removing heat from it at a rate of 360 kJ/min. If the required power input to the refrigerator is 2 kW, determine (i) the C.O.P. of the refrigerator and (ii) the rate of heat rejection to the room that houses the refrigerator. Also state the Kelvin Planck and Clausius statement being used for second law of thermodynamics. Further define C.O.P. of refrigerator, C.O.P. of heat pump and show that $(C.O.P.)_{\text{heat pump}} = 1 + (C.O.P.)_{\text{refrigerator}}$

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34. [UNKNOWN] paper-ii

8. (a) The internal energy of a certain substance is expressed by the equation $u = 3.62 pv + 86$ where u is in kJ/kg, p in kPa and v is in m³/kg. A system composed of 5 kg of this substance expands from an initial pressure of 550 kPa and a volume of 0.25 m³/kg to a final pressure of 125 kPa, in a process in which pressure and volume are related by $pv^{1.2} = \text{constant}$. If the expansion process is quasistatic, determine work (W), change in internal energy and heat transferred in this process.

10 The performance of an air-conditioner unit rated as 40 Tons, seems to be indicating poor cooling. The test on heat rejection to atmosphere in its condenser shows the following: Cooling water flow : 4 L/s Water inlet temperature : 30°C Water outlet temperature : 40°C Power input to motor : 40 kW Efficiency of motor : 95% Specific heat of water : 4.186 kJ/kg K Calculate the actual refrigerating capacity of the unit and conclude on whether the unit is giving poor cooling.

10 A four-cylinder S.I. engine has a bore of 60 mm and a stroke of 85 mm. It runs at 3000 rpm and is tested at this speed against a brake which has a torque arm of 0.35 m. The net brake load is 160 N and the fuel consumption is 6.6 lit/hr. The specific gravity of the fuel used is 0.78 and it has a lower calorific value of 44,000 kJ/kg. A Morse-test is carried out and the cylinders are cut out in the order 1, 2, 3, 4 with the corresponding brake loads of 114, 110, 112 and 116N respectively. Calculate for this speed the bp, the bmep, the bte, the bsfc, the ip, the mechanical efficiency and the imep.

20 The temperature distribution across a wall, having thickness of 1 m, at an instant of time is given as : $T(x) = 900 - 300x - 50x^2$ where T is in degree celsius and x is in metres. The uniform heat generation of 1000 W/m³ is present in the wall of area 10 m² having density $\rho = 1600$ kg/m³, thermal conductivity $k = 40$ W/mK and specific heat $C = 4$ kJ/kg K. Determine (i) the rate of heat transfer entering the wall and leaving the wall. the rate change of internal energy of the wall. the time rate of temperature change at $x = 0$ and at $x = 0.5$ m.

20 Steam is supplied by the steam generator at 90 bar and 500°C. After the expansion in the turbine to 10 bar a portion of the steam is bled for regenerative feed heating and the remaining is passed on to the reheater from where it returns to the turbine at 500°C. Expansion further continues to 0.07 bar. For 1 kg of mass of steam supplied at generator inlet, calculate (a) heat supplied, (b) heat rejected, (c) net work done, (d) thermal efficiency, and (e) steam rate in kg/kWh. Assume the specific volume of water as 0.001074 m³/kg. Specific enthalpy of saturated liquid at 0.07 bar and 10 bar are 163.38 kJ/kg and 762.61 kJ/kg respectively.

20 7 fsi-d-mche ===== PAGE 8 ===== pp O — 0 o CV MOWER CHART lthr41Adre4ite The handle of a saucepan, 30 cm long and 2 cm in diameter is partially immersed in boiling water at 100°C. The average unit conductance over the handle surface is 7.35 W/m² K in the kitchen air at 24°C. The cook is likely to grasp the last 10 cm of the handle and hence, the temperature of this portion should not exceed 32°C. What should be the material conductivity of handle? The handle may be treated as a fin of insulated tip.

10 An artificial spherical satellite flies around the earth. Calculate the temperature of the satellite surface, assuming that there are no heat sources and surface temperature is uniform all over the surface. The solar radiation reflected from the earth and radiation emitted from the earth should also be ignored. If $\alpha_s = 0.2$ and $E_s = 1$ If surface of the satellite is gray Find the ratio ϵ_s/α_s , when the temperature of the satellite surface becomes 30°C. The incident solar radiation is 1500 W/m².

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35. [UNKNOWN] paper-ii

15 - 20 cK as \ DRY BULB TEMPERATURE, °C \ 30 35 -40 50 max" 0.75 re/kg oao m³/kg • m³/kg 0.90 m³/kg BELOW 0 °C PROPERTIES AND ENTHALPY 0.85 DEVIATION LINES ARE FOR ICE SPECIFIC VOLUME, m³/kg DRY AIR PSYCHROMETRIC CHART BAROMETRIC PRESSURE 1.01325 bar SEA LEVEL Do not write your Roll No. on this Sheet c>, 4.0 120 125 120 135

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36. [UNKNOWN] paper-ii

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37. [UNKNOWN] paper-ii

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